

Git Commands: -M and -u

Topic: What git branch -M main and git push -u origin main mean.

1. git branch -M main

Purpose: Rename the current Git branch to **main**.

Breakdown:

- git branch = work with branches
- -M = force a branch rename
- main = the new branch name

```
git branch -M main
```

Think: **-M = rename my branch to main**. It is the stronger/forced form of -m.

2. git push -u origin main

Purpose: Upload your local **main** branch to the remote repository and remember that remote branch as its upstream.

Breakdown:

- git push = send local commits to a remote repository
- origin = the usual name for your remote repository (for example, GitHub)
- main = the branch being pushed
- -u = set the upstream/tracking branch

```
git push -u origin main
```

3. What does -u really do?

It tells Git: **"Remember that my local main branch tracks origin/main."**

Because Git remembers this connection, later you can usually use:

```
git push  
git pull
```

instead of repeatedly writing git push origin main and git pull origin main.

4. Why are the two commands often used together?

```
git branch -M main
```

Rename the current branch to **main**.

```
git push -u origin main
```

Push **main** to the remote and remember the remote branch as its upstream.

5. Simple analogy

Your local Git branch is like your **notebook**. GitHub is like the **cloud copy**.

```
git push -u origin main
```

-u is like saying: "This notebook belongs to that cloud copy. Remember this connection."

6. Quick reference

Part	Meaning
-M	Force/ensure branch rename
main	New branch name
push	Upload commits to a remote
origin	Remote repository nickname
-u	Set and remember the upstream branch

7. Memory hook

-M -> Make/rename the branch.

-u -> Upstream: remember where this branch tracks.

Key takeaway: git branch -M main changes the branch name; git push -u origin main pushes it to the remote and sets the tracking connection.